



Technical Service Bulletin

Technical Service Bulletin: tsb020001

Released Date: 09-Oct-2003

CENSE™ Phase 1.5 Harness Repair Instructions

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Core Issue

It has recently been discovered that the all Phase 1.5 CENSE™ right-bank wiring harness, Part Number 4017464, on QSK60C engines have been produced incorrectly. The exhaust gas temperature connectors have been wired incorrectly for power cylinders 1, 2, 5, and 7, resulting in incorrect diagnosis of cylinder fault codes.

Confirmation

Engine Serial Numbers Affected (QSK60 Only)							
33150763	33150602	33150387	33150015	33149343	33148906	33148449	33147903
33150758	33150598	33150374	33149988	33149314	33148901	33148444	33147878
33150755	33150596	33150363	33149973	33149277	33148897	33148401	33147584
33150710	33150591	33150345	33149964	33149264	33148890	33148390	33147332
33150706	33150587	33150341	33149951	33149258	33148781	33148385	33147260
33150695	33150585	33150337	33149934	33149252	33148684	33148377	33147083
33150687	33150580	33150331	33149903	33149247	33148620	33148355	33147075
33150674	33150576	33150321	33149719	33149204	33148616	33148304	33146876
33150668	33150573	33150313	33149702	33149201	33148608	33148280	33146847

Engine Serial Numbers Affected (QSK60 Only)							
3315066 6	3315056 8	3315030 5	3314968 8	3314915 9	3314860 1	3314825 9	3314681 9
3315065 7	3315056 3	3315029 1	3314964 0	3314907 7	3314859 1	3314824 4	3314680 9
3315065 1	3315056 1	3315027 0	3314949 3	3314907 5	3314858 6	3314821 5	
3315064 9	3315055 7	3315026 2	3314947 1	3314902 9	3314853 9	3314819 6	
3315064 5	3315052 2	3315018 9	3314946 1	3314896 8	3314853 4	3314818 2	
3315063 5	3315051 8	3315018 1	3314942 1	3314896 2	3314852 8	3314817 0	
3315062 5	3315042 0	3315016 9	3314940 8	3314895 4	3314846 7	3314812 2	
3315061 9	3315040 7	3315014 7	3314935 9	3314894 8	3314846 3	3314807 7	
3315061 5	3315039 7	3315013 4	3314935 6	3314894 0	3314845 6	3314800 9	

Note : Subsequent engines from and including ESN 33146793 were rectified in the manufacturing plant and therefore require no modification.

As a result, the temperature readings and fault codes obtained via the CENSE™ diagnostic software, INSITE™, do **not** correspond correctly to their associated power cylinders.

For example, INSITE™ can indicate low or high exhaust gas temperature from cylinder number 1 when it is actually measuring the exhaust gas from cylinder number 7. As a consequence, this would mislead a technician into disassembling a healthy power cylinder when investigating the cause of the fault code.

This is a confirmed issue with the Phase 1.5 CENSE™ harness; therefore, there is no requirement for any inspection or fault verification procedure.

The cause of this issue is through a wiring harness design error.

Mis-diagnosing power cylinder performance can lead an unhealthy power cylinder going undetected. This would obviously result in degradation in general engine performance and possible progressive component damage.

Resolution

Due to un-serviceability of the right bank CENSE™ harness (in which the fault lies), the repair instructions instruct modification to the wiring within the CENSE™ ECM housing. Altering the wiring at this location essentially “cancels out” the mis-wire on the right bank harness.

GRAPHIC NOT FOUND

Remove the six M6 capscrews.

Remove the CENSE™ housing cover.

GRAPHIC NOT FOUND

Remove the module upper connector with a 4-mm allen key as shown.

GRAPHIC NOT FOUND

Using the Deutsch pin removal tool, Part Number 3824815, remove and reinsert the pins into their correct terminal. See table 1.

Label each wire when disconnected with the new pin location.

As each pin is reinserted, there will be a click as the terminal locks in the connector. Give the wire a gentle pull to ensure it is located.

Table 1	
Remove from Terminal	Reinsert into Terminal
31	01
21	11
01	14
11	04
14	22
04	32
22	31
32	21

GRAPHIC NOT FOUND

Reconnect the connector with the 4-mm allen key.

Torque Value: 3 n•m [27 in-lb]

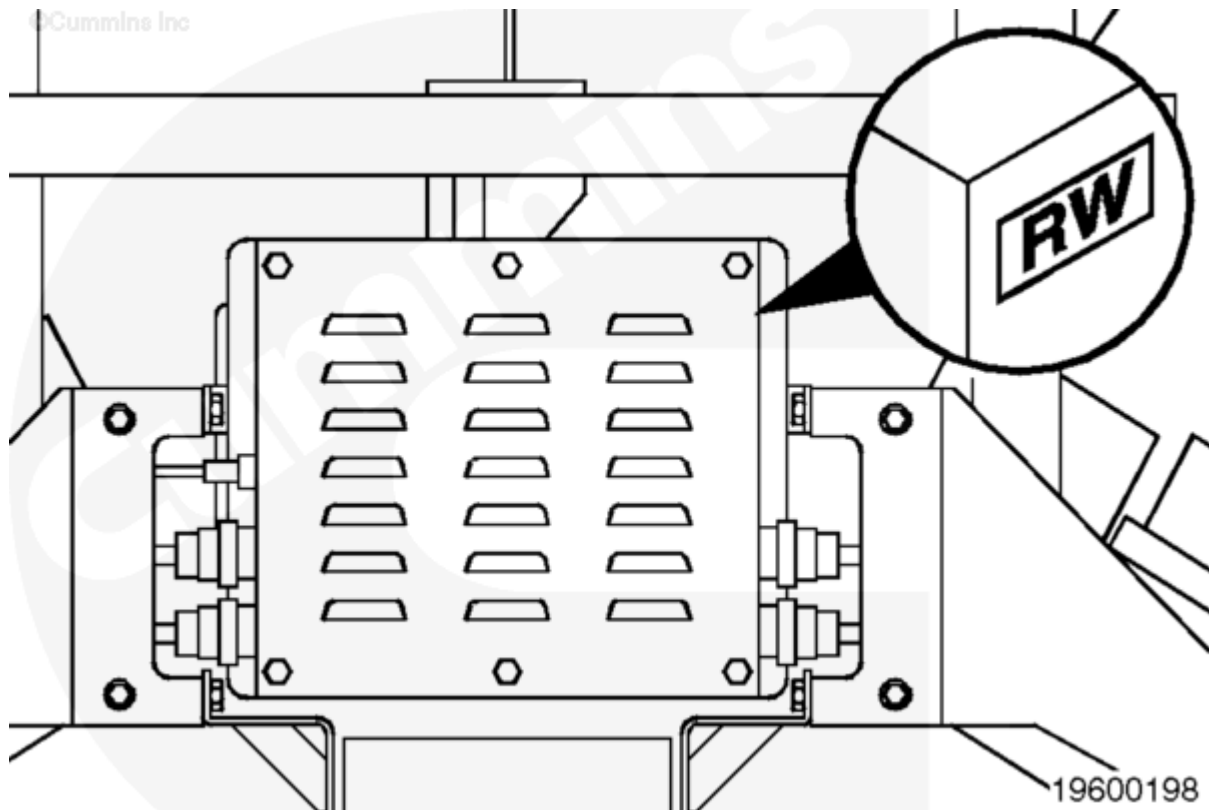
GRAPHIC NOT FOUND

Refit the CENSE™ housing cover.

Torque Value: 10 n•m [89 in-lb]

After completing the rewire, plug the connector back into the CENSE™ ECM and test using CENSE™ INSITE™. Do this by going into the real time monitor feature, selecting all 16 EGT sensors. Disconnect the exhaust thermocouples from the wiring harness one at a time while observing the real time monitor display. Ensure that the correct cylinder temperature reading goes to “failed” as its associated sensor is disconnected. If everything is functioning correctly, move onto the next sensor.

Once satisfied that the CENSE™ system is functioning correctly, there is a need to identify that the CENSE™ wiring has been modified. This **must** be done by stamping the word “RW” (rewired) with letter stamps on the right hand side of the CENSE™ ECM housing.



Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.